WORKSHEET

Chapter 2: Complex number conversions

14	1	7	11	9		13	14	12	6	10	3	12		11	5	12	14	10	8				
																				11	4	7	
																				13	10	4	12	6	2	7	9	9

Harvey MacKay

Questions

For $z = -3 + 3i$, calculate:

1 $|z|$

2 θ

3 Write z in polar form.

For $z = 2\sqrt{3} + 2i$, calculate:

4 $|z|$

5 θ

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6 Write z in polar form.

Convert the following vectors to polar form.

7 $(3, -5)$

8 $\begin{pmatrix} -7\sqrt{3} \\ 7 \end{pmatrix}$

Write the following complex numbers in Cartesian form:

9 $5 \operatorname{cis}\left(\frac{\pi}{3}\right)$

10 $7\sqrt{2} \operatorname{cis}\left(\frac{3\pi}{4}\right)$

11 $4 \operatorname{cis}\left(-\frac{\pi}{6}\right)$

12 $2\sqrt{2} \operatorname{cis}\left(-\frac{3\pi}{4}\right)$

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Convert the follow vectors into Cartesian form:

13 $\left(10, \frac{2\pi}{3}\right)$

14 $\left(2\sqrt{17}, -14^\circ\right)$

Solutions

A $2\sqrt{3} - 2i$

C $\frac{\pi}{6}$

D $3\sqrt{2}$

E $\sqrt{34}\,\text{cis}(-59°)$

H $4\,\text{cis}\left(\frac{\pi}{6}\right)$

I $8 - 2i$

L $\frac{3\pi}{4}$

N $14\,\text{cis}\left(\frac{5\pi}{6}\right)$

O $-7 + 7i$

R 4

S $\frac{5}{2} + \frac{5\sqrt{3}}{2}i$

T $-2 - 2i$

U $3\sqrt{2}\,\text{cis}\left(\frac{\pi}{4}\right)$

W $-5 + 5\sqrt{3}\,i$

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